

Question Paper

Exam Date & Time: 22-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Introduction to internet of things [NPT_4129]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss the IoT protocols that you recommend the most suitable connectivity solution for an industrial monitoring system with high interference and long-range requirements. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Analyse the differences between CoAP and XMPP (5)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Discuss the Integration of Sensors and Actuators with Arduino with a case of your choice. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) Suppose you are developing a health-monitoring Sensor-Cloud for remote patient care. What sensors and actuators would you include? Apply your understanding to write the workflow in detail. (5)
Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) A self-driving car uses multiple IoT-enabled cameras mounted at different angles. Describe the data flow from sensors to the cloud and how it supports autonomous decision-making. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Analyse the concept of **V2X communication** to design a scenario where vehicles cooperate to manage a highway merge automatically. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyse your understanding of connected vehicle systems to describe how sensor integration and IoT connectivity improve driver assistance functions (5)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
B) Apply the reliability and scalability requirements of IIoT to suggest suitable network architectures for continuous industrial operations. (5)
Course Outcome: "CO4" / Cognitive Level: "Application"
- 5) Analyze the concept of IoT to design a smart irrigation system that optimizes water usage in agriculture. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

B)

Apply your understanding of IoT architectures to compare how data collection, communication, and analytics differ across the three domains - Agriculture, Healthcare, and Activity Monitoring. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4129:Introduction to internet of things

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4129:Introduction to internet of things

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)